

Lecture 13

EE-215 Electronic Devices and Circuits

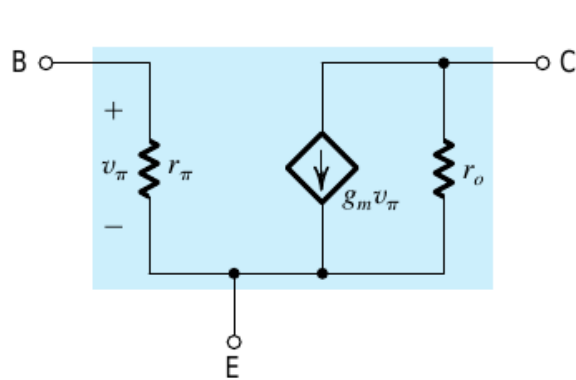
Asst Prof Muhammad Anis Ch

Basic BJT Amplifier Configurations

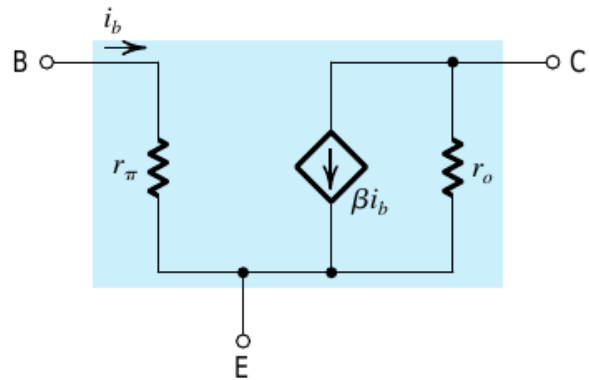
Table 6.4 Small-Signal Models of the BJT

Hybrid- π Model

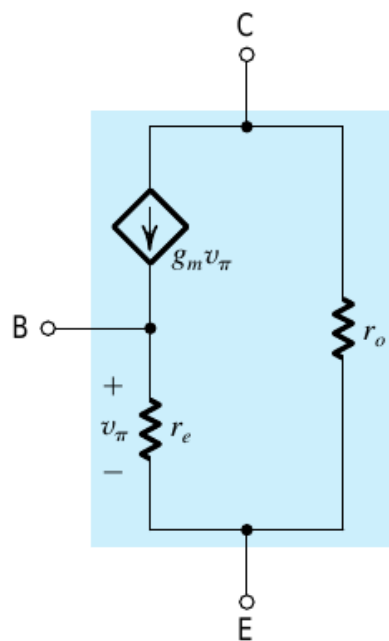
■ $(g_m v_\pi)$ Version



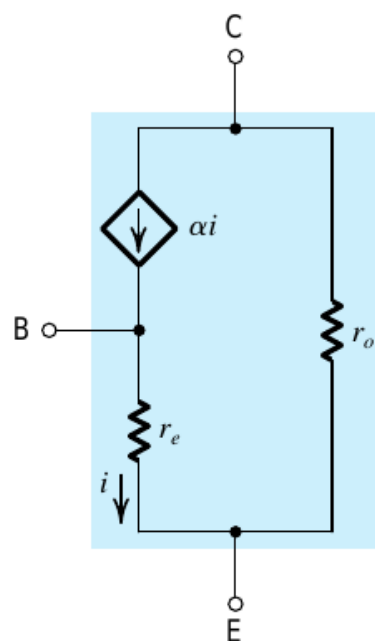
■ (βi_b) Version



■ $(g_m v_\pi)$ Version

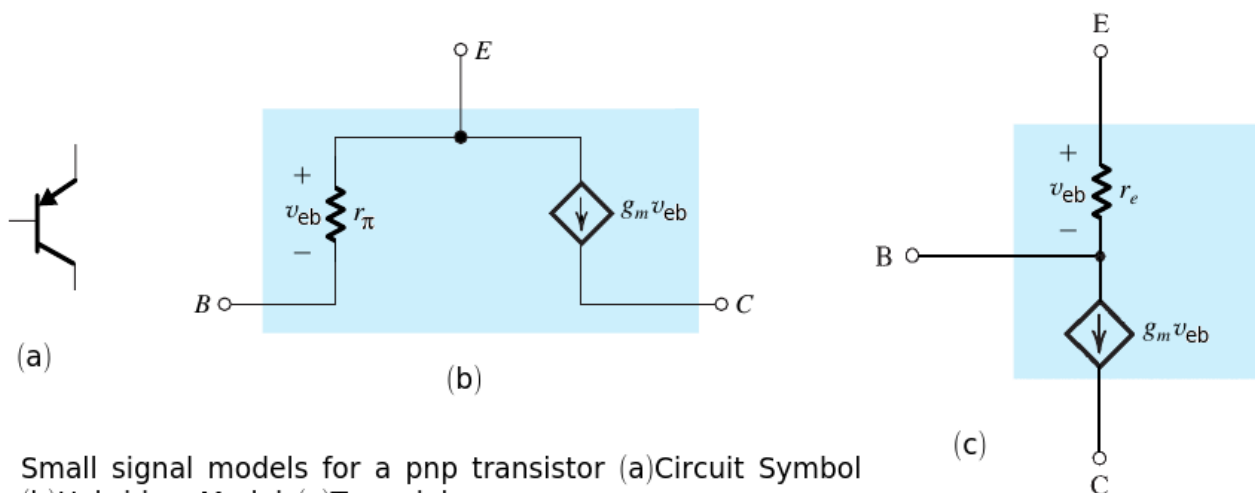


■ (αi) Version



Model Parameters in Terms of DC Bias Currents			
$g_m = \frac{I_C}{V_T}$	$r_e = \frac{V_T}{I_E} = \alpha \frac{V_T}{I_C}$	$r_\pi = \frac{V_T}{I_B} = \beta \frac{V_T}{I_C}$	$r_o = \frac{ V_A }{I_C}$
In Terms of g_m			
$r_e = \frac{\alpha}{g_m}$	$r_\pi = \frac{\beta}{g_m}$		
In Terms of r_e			
$g_m = \frac{\alpha}{r_e}$	$r_\pi = (\beta + 1)r_e$	$g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$	
Relationships between α and β			
$\beta = \frac{\alpha}{1 - \alpha}$	$\alpha = \frac{\beta}{\beta + 1}$	$\beta + 1 = \frac{1}{1 - \alpha}$	

- Note that, this table applies equally well to both the npn and pnp transistors with no change in polarities
- for a pnp transistor,

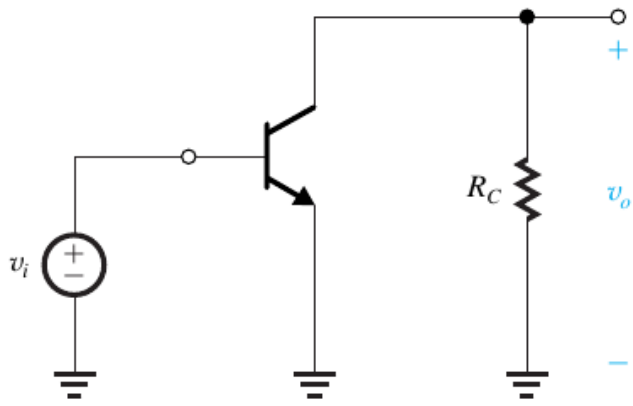


Small signal models for a pnp transistor (a)Circuit Symbol
(b)Hybrid- π Model (c)T model

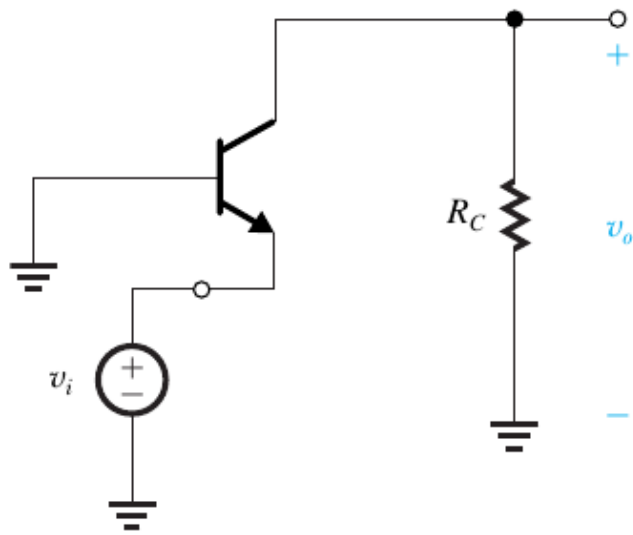
The base-width modulation effect can be included by placing r_o between C and E

The Three Basic Configurations

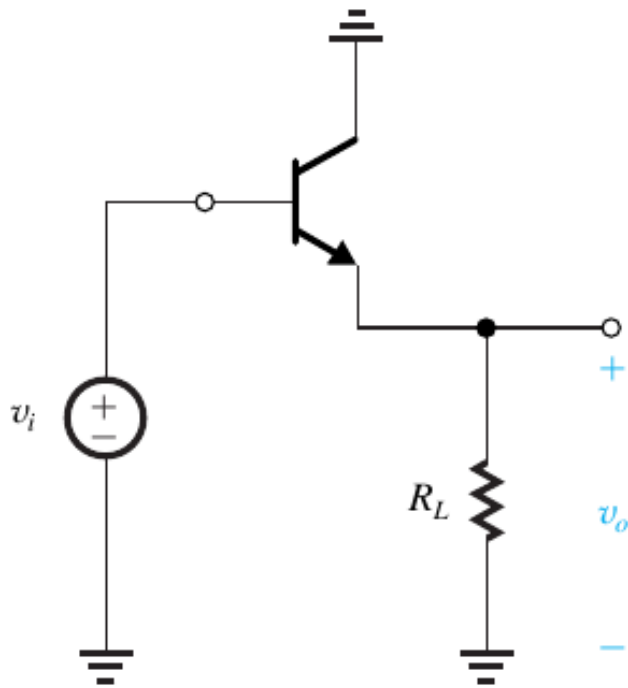
- There are three basic configurations for connecting the BJT as an amplifier, namely
 - the Common Emitter (CE) Amplifier
 - the Common Base (CB) Amplifier
 - The Common Collector (CC) Amplifier (also called Emitter Follower)
 - Each of these configurations is obtained by
 - connecting one of the three BJT terminals to ground
 - thus creating a two-port network with the grounded terminal
 - being common to the input and output ports
 - The resulting three configurations with biasing arrangement omitted are shown in fig



(a) Common-Emitter (CE)



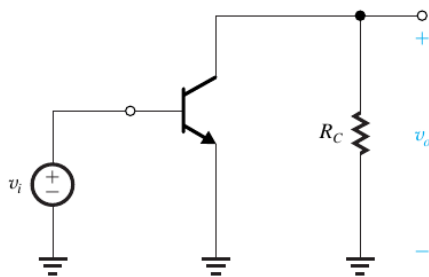
(b) Common-Base (CB)



(c) Common-Collector (CC)
or Emitter Follower

Common Emitter Amplifier

- here the emitter terminal is connected to ground,
 - the input voltage signal is applied at the base (w.r.t ground),
 - the output voltage signal is taken at the collector (w.r.t ground)

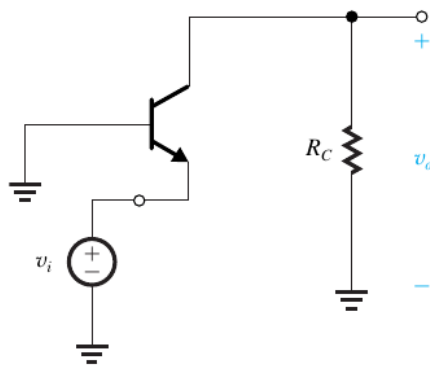


(a) Common-Emitter (CE)

- this configuration is called the grounded emitter or the common emitter (CE) amplifier
- Note that dc biasing circuit is not shown in figure

Common Base Amplifier

- The common base (CB) or grounded base amplifier is obtained
 - by connecting the base to ground,
 - applying the input between the emitter and ground
 - taking the output v_o between the collector and ground

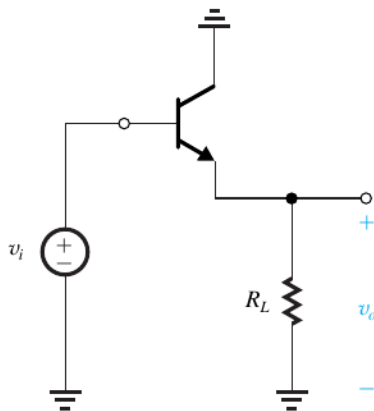


(b) Common-Base (CB)

- Note that dc biasing circuit is not shown in figure

Common Collector Amplifier

- The common collector (CC) or grounded collector amplifier is obtained
 - by connecting the collector terminal to ground,
 - applying the input between the base and ground
 - taking the output v_o between the emitter and ground

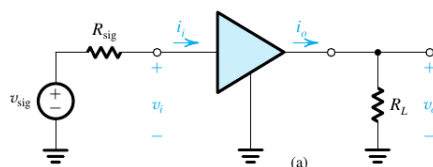


(c) Common-Collector (CC)
or Emitter Follower

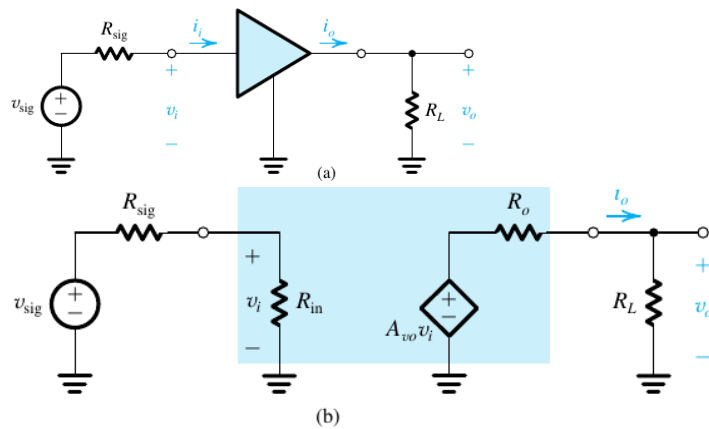
- Note that dc biasing circuit is not shown in figure
- the Common Collector amplifier is also called Emitter Follower

Characterizing Amplifiers

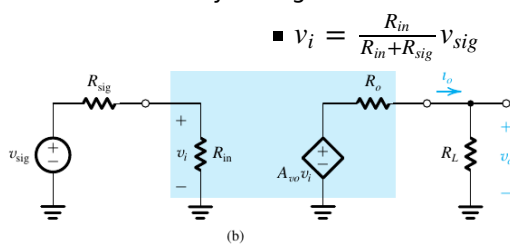
- lets take a look at how to characterize the performance of an amplifier as a circuit building block
 - here an amplifier is fed with a signal source
 - having an open-circuit voltage v_{sig} and an internal resistance R_{sig}



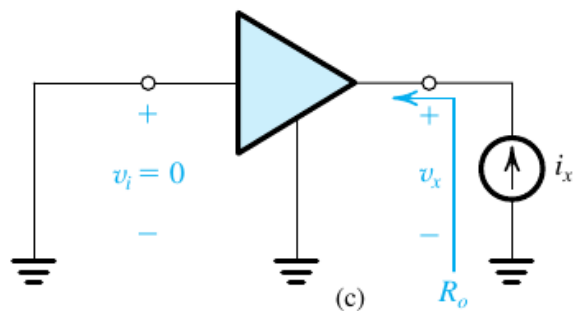
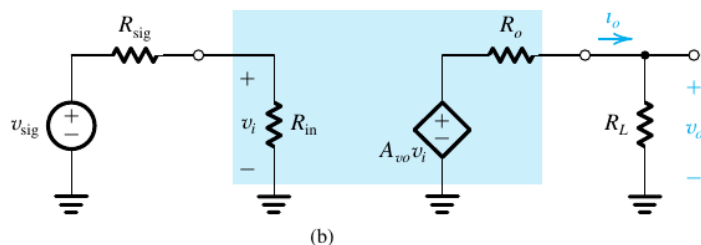
- this voltage source (v_{sig} , R_{sig}) can be an actual signal source
 - or in a cascade amplifier, it can be the Thevenin equivalent of the output circuit of the preceeding amplifier.
- the load resistor R_L is connected at the output terminal of the amplifier
 - this R_L can be an actual load resistor
 - or the input resistance of a succeeding amplifier stage in a cascade amplifier
- The amplifier block can be replaced by its equivalent circuit model



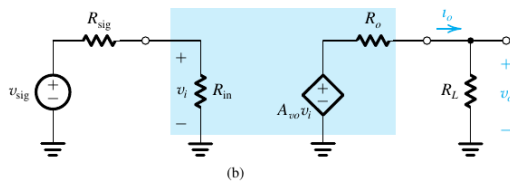
- the input resistance R_{in}
 - indicate the loading effect of the amplifier input on the signal source
 - and is given as $R_{in} = \frac{v_i}{i_i}$
- by voltage divider rule



- $v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig}$
- A_{vo} is the open-circuit voltage gain and is defined as
 - $A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L = \infty}$



- the 3rd parameter that is essential to characterize an amplifier is the output resistance R_o
 - R_o is the resistance seen looking back into the amplifier output terminal with v_i set to zero
 - $\Rightarrow R_o = \frac{v_x}{i_x}$
 - Note that the controlled source $A_{vo} v_i$ and the output resistance R_o
 - represent the Thevenin equivalent of the amplifier output circuit.



- by voltage divider
 - the output voltage v_o can be given as

$$v_o = \frac{R_L}{R_L + R_o} (A_{vo} v_i) \Rightarrow \frac{v_o}{v_i} = A_{vo} \frac{R_L}{R_L + R_o}$$
 - thus the voltage gain of the amplifier is

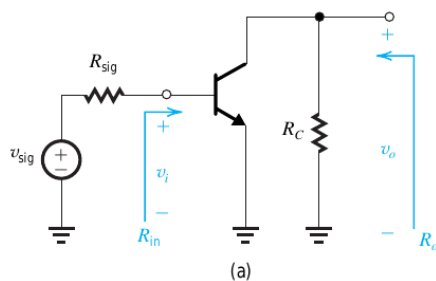
$$A_v = \frac{v_o}{v_i} = A_{vo} \frac{R_L}{R_L + R_o}$$
 - the overall voltage gain is

$$G_v = \frac{v_o}{v_{sig}} = A_{vo} \frac{R_L}{R_L + R_o} \frac{v_i}{v_{sig}}$$

$$G_v = A_{vo} \frac{R_L}{R_L + R_o} \frac{R_{in}}{R_{in} + R_{sig}} \because v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig}$$

The Common-Emitter (CE) Amplifier

- is the most widely used BJT amplifier configuration



- Note that, here the biasing arrangement is not shown (the circuit in fig is the ac equivalent circuit)
- the signal source has an internal resistance of R_{sig} and open circuit voltage of v_{sig}
- let's analyze this circuit to determine R_{in} , A_{vo} , R_o , A_v and G_v
- Note that R_C is part of the amplifier circuit, and when a load resistance R_L is connected to the amplifier output, it appears in parallel with R_C
- replacing BJT with its hybrid- π model

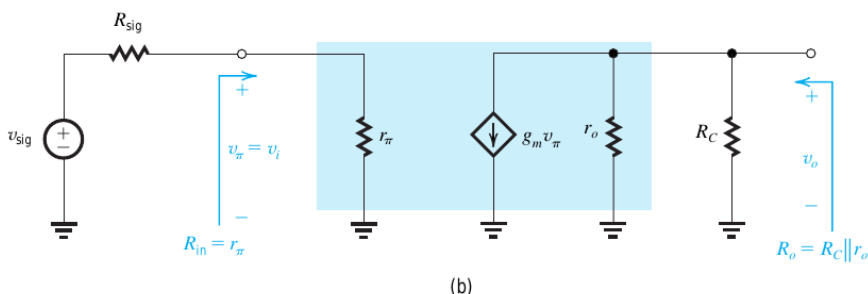
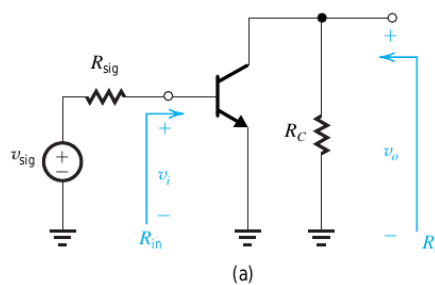
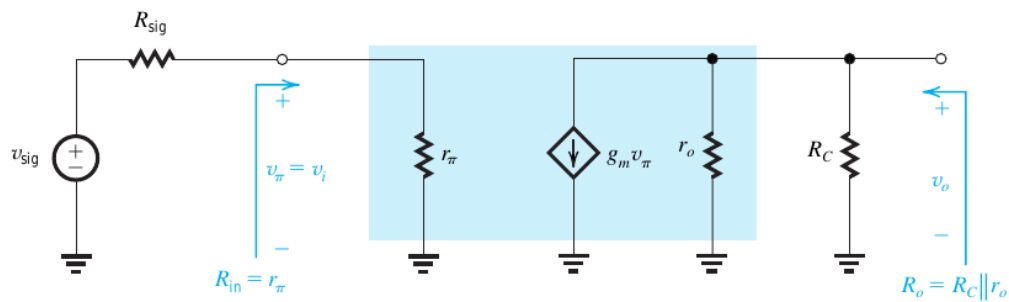


Figure 6.50 (a) Common-emitter amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig} .

- (b) The common-emitter amplifier circuit with the BJT replaced with its hybrid- π model.



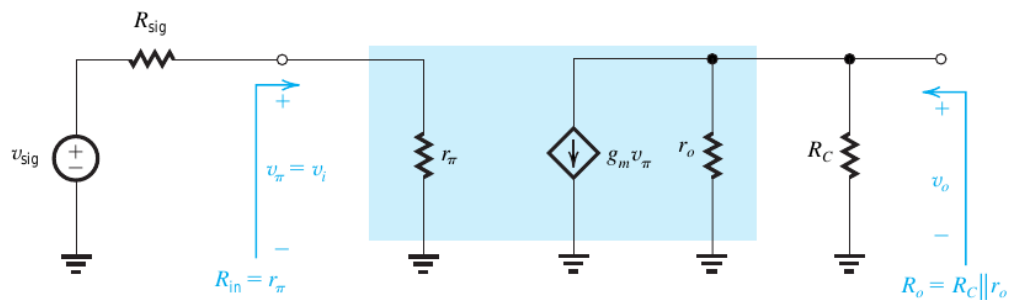
(b)

- this equivalent circuit can be used to determine the characteristic parameters (R_{in} , A_{vo} , R_o) of the amplifier
 - from the fig

$$\blacksquare \Rightarrow \text{input resistance} = R_{in} = \frac{v_i}{i_b} = r_{\pi}$$

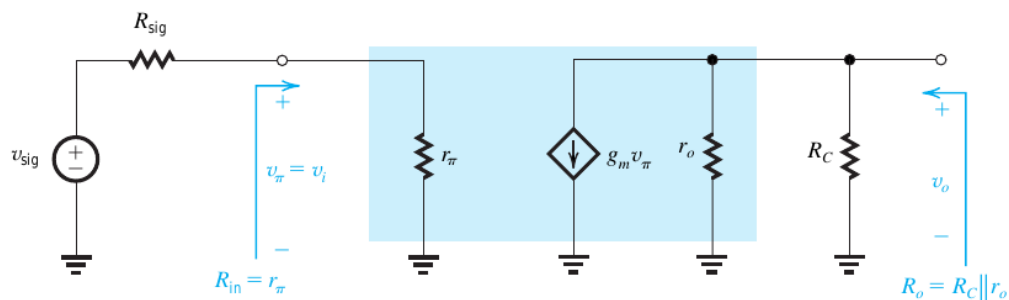
$$\circ \text{ by KCL} \Rightarrow g_m v_{\pi} + \frac{v_o}{r_o \parallel R_C} = 0$$

$$\blacksquare v_o = -g_m v_{\pi} (r_o \parallel R_C) \Rightarrow \frac{v_o}{v_{\pi}} = -g_m (r_o \parallel R_C)$$



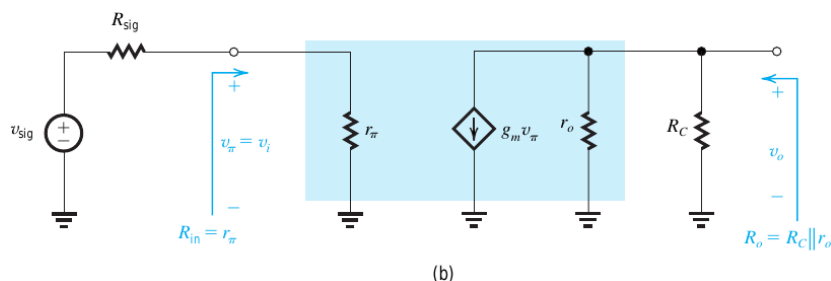
(b)

- $\frac{v_o}{v_{\pi}} = -g_m (r_o \parallel R_C)$
 - here $v_i = v_{\pi} \Rightarrow A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L=\infty} = \left. \frac{v_o}{v_{\pi}} \right|_{R_L=\infty} = -g_m (r_o \parallel R_C)$
 - if the base-width modulation can be ignored $r_o \rightarrow \infty$ ($r_o > R_C$)
 - $\blacksquare \Rightarrow A_{vo} \approx -g_m R_C$

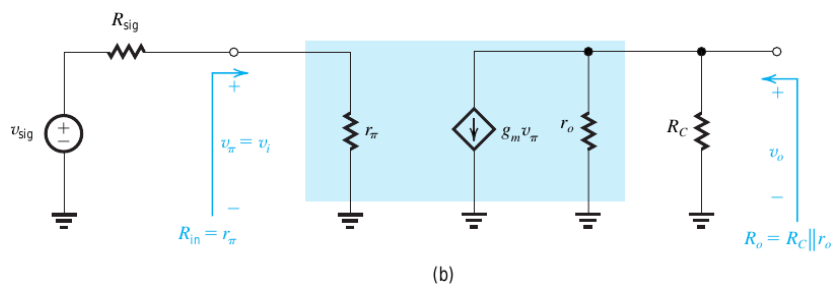


(b)

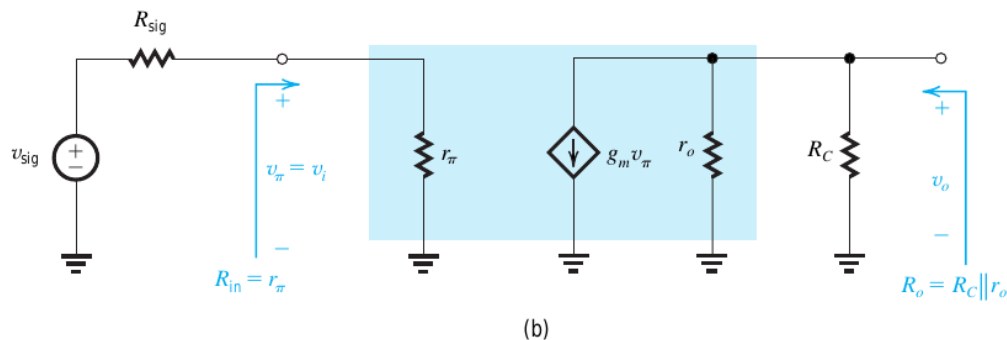
- Now the output resistance R_o is the resistance seen looking into the output terminal with v_i set to zero
 - $\blacksquare v_i = 0 \Rightarrow v_{\pi} = 0 (\because v_i = v_{\pi}) \Rightarrow g_m v_{\pi} = 0$
 - $\blacksquare \Rightarrow R_o = R_C \parallel r_o$
 - thus for CE amplifier
 - $\blacksquare R_{in} = r_{\pi}, R_o = R_C \parallel r_o, A_{vo} = -g_m (r_o \parallel R_C)$



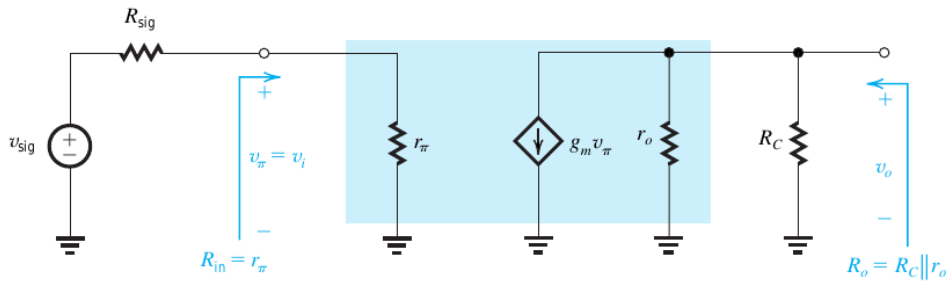
- - now to determine A_v , the load resistor R_L is connected
 - i.e. $A_v = \frac{v_o}{v_i}$
 - by KCL
 - $\frac{v_o}{(R_C || r_o || R_L)} = -g_m v_\pi$
 - $v_o = (-g_m v_\pi)(R_C || r_o || R_L)$
 - $\Rightarrow A_v = \frac{v_o}{v_i} = \frac{v_o}{v_\pi} = -g_m (R_C || r_o || R_L)$



- - now the overall voltage gain G_v can be determined as follows
 - $G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v$
 - As $A_v = \frac{v_o}{v_i} = \frac{v_o}{v_\pi} = -g_m (R_C || r_o || R_L)$ and by voltage divider rule, $v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig}$
 - As $R_{in} = r_\pi \Rightarrow \frac{v_i}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} = \frac{r_\pi}{r_\pi + R_{sig}}$
 - $\Rightarrow G_v = \frac{v_i}{v_{sig}} A_v = \frac{r_\pi}{r_\pi + R_{sig}} \left\{ -g_m (R_C || r_o || R_L) \right\}$
 - Note that A_v , G_v are determined with load resistor R_L connected

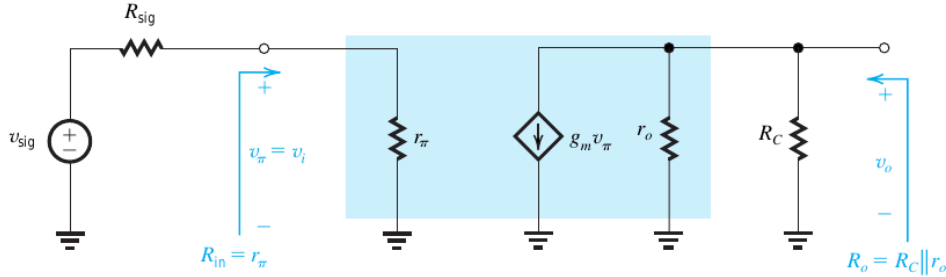


- - $A_v = -g_m (R_C || r_o || R_L)$, $G_v = \frac{r_\pi}{r_\pi + R_{sig}} \left\{ -g_m (R_C || r_o || R_L) \right\}$
 - A_v and G_v can also be expressed in terms of total resistance in collector and total resistance in base, as follows
 - As $r_\pi = \frac{\beta}{g_m}$, $r_e = \frac{\alpha}{g_m} \Rightarrow g_m = \frac{\beta}{r_\pi} = \frac{\alpha}{r_e}$
 - $A_v = -g_m (R_C || r_o || R_L)$, $G_v = \frac{r_\pi}{r_\pi + R_{sig}} \left\{ -g_m (R_C || r_o || R_L) \right\}$



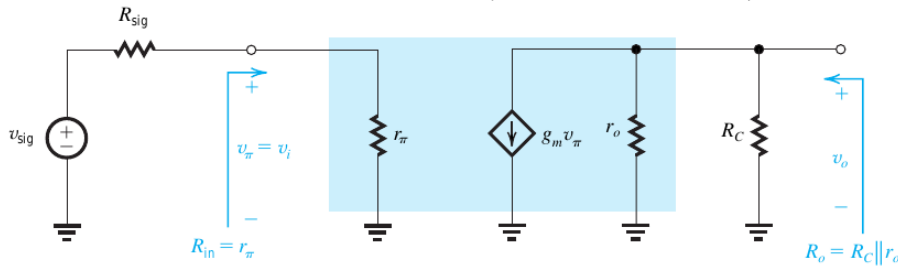
(b)

- - $g_m = \frac{\beta}{r_\pi}$
 - $\Rightarrow A_v = -\frac{\beta}{r_\pi} (R_C \parallel r_o \parallel R_L) = -\beta \frac{(R_C \parallel r_o \parallel R_L)}{r_\pi}$
 - $\Rightarrow A_v = -\beta \frac{\text{Total resistance in collector}}{\text{Total resistance in base}}$
 - $g_m = \frac{\alpha}{r_e}$
 - $\Rightarrow A_v = -\frac{\alpha}{r_e} (R_C \parallel r_o \parallel R_L) = -\alpha \frac{(R_C \parallel r_o \parallel R_L)}{r_e}$
 - $\Rightarrow A_v = -\alpha \frac{\text{Total resistance in collector}}{\text{Total resistance in emitter}}$
- $A_v = -g_m (R_C \parallel r_o \parallel R_L), G_v = \frac{r_\pi}{r_\pi + R_{sig}} \left\{ -g_m (R_C \parallel r_o \parallel R_L) \right\}$



(b)

- - $g_m = \frac{\beta}{r_\pi}$
 - $G_v = \frac{r_\pi}{r_\pi + R_{sig}} \left\{ -\frac{\beta}{r_\pi} (R_C \parallel r_o \parallel R_L) \right\} = \frac{1}{r_\pi + R_{sig}} \left\{ -\beta (R_C \parallel r_o \parallel R_L) \right\} = -\beta \frac{(R_C \parallel r_o \parallel R_L)}{r_\pi + R_{sig}}$
 - $G_v = -\beta \frac{(R_C \parallel r_o \parallel R_L)}{r_\pi + R_{sig}} = -\beta \frac{\text{Total resistance in collector}}{\text{Total resistance in base}}$
- $A_v = -g_m (R_C \parallel r_o \parallel R_L), G_v = \frac{r_\pi}{r_\pi + R_{sig}} \left\{ -g_m (R_C \parallel r_o \parallel R_L) \right\}$

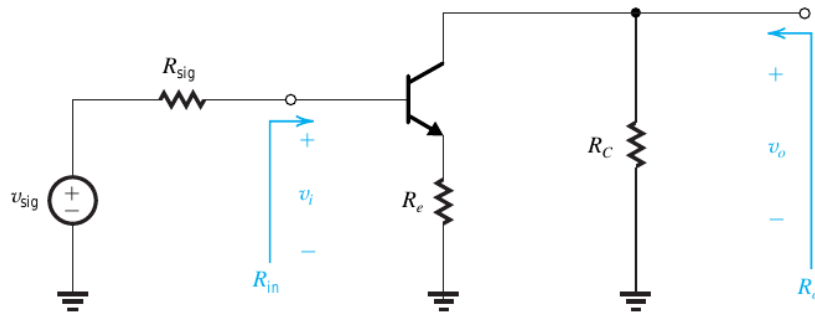


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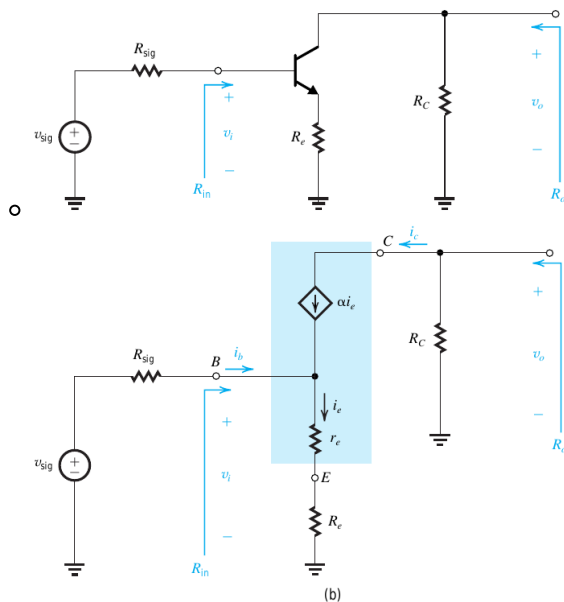
- - $g_m = \frac{\alpha}{r_e} \Rightarrow G_v = \frac{r_\pi}{r_\pi + R_{sig}} \left\{ -\frac{\alpha}{r_e} (R_C \parallel r_o \parallel R_L) \right\}$
 - $G_v = \frac{(\beta+1)r_e}{(\beta+1)r_e + R_{sig}} \left\{ -\frac{\alpha}{r_e} (R_C \parallel r_o \parallel R_L) \right\} = \frac{(\beta+1)}{(\beta+1)r_e + R_{sig}} \left\{ -\alpha (R_C \parallel r_o \parallel R_L) \right\}$
 - As $r_\pi = (\beta+1)r_e$
 - $G_v = \frac{1}{\frac{(\beta+1)r_e + R_{sig}}{(\beta+1)}} \left\{ -\alpha (R_C \parallel r_o \parallel R_L) \right\} = \frac{-\alpha (R_C \parallel r_o \parallel R_L)}{r_e + \frac{R_{sig}}{(\beta+1)}} = -\alpha \frac{(R_C \parallel r_o \parallel R_L)}{r_e + R_{sig}/(\beta+1)}$

$$\blacksquare G_v = -\alpha \frac{(R_C \parallel r_o \parallel R_L)}{r_e + R_{sig} / (\beta + 1)} = -\alpha \frac{\text{Total resistance in collector}}{\text{Total resistance in emitter}}$$

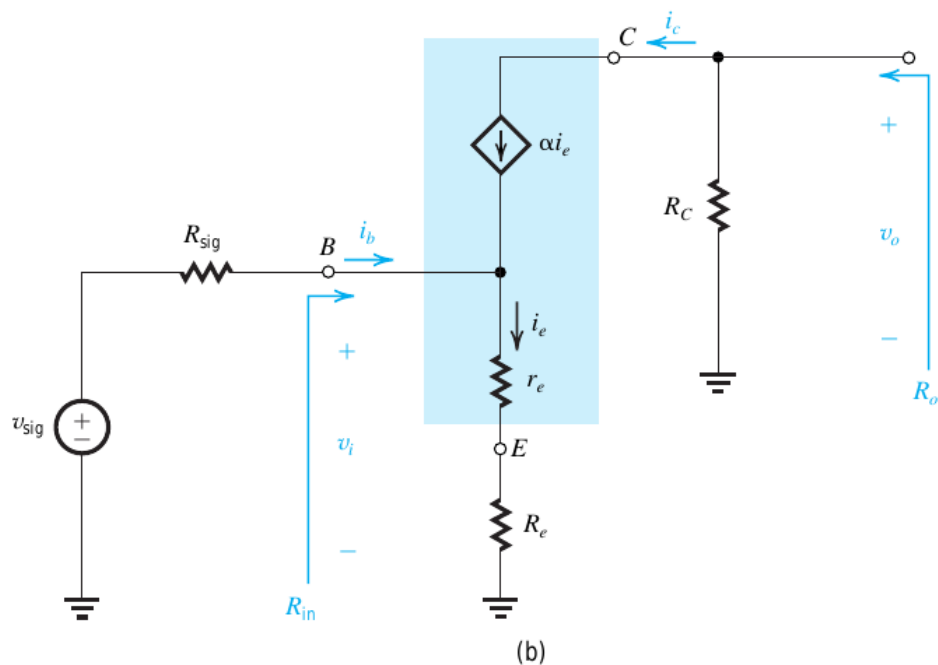
The Common-Emitter Amplifier with an Emitter Resistance



- o here we have placed a resistance R_e in the emitter lead of the common-emitter amplifier
 - o the resistance R_e can lead to significant changes in amplifier characteristics
 - o thus R_e provide extra flexibility to adjust the amplifier characteristics to fit the design requirements
 - o ignoring base-width modulation effect (i.e. ignoring Early effect)
- o the small-signal equivalent circuit can be drawn as



- o here T-model is preferred over the π model, because it makes the analysis somewhat simple
 - o In general, whenever a resistance is connected in the emitter lead, the T model is preferred.



- As a consequence of the T-model, R_e appears in series with the resistance r_e
- to determine the amplifier input resistance, R_{in}

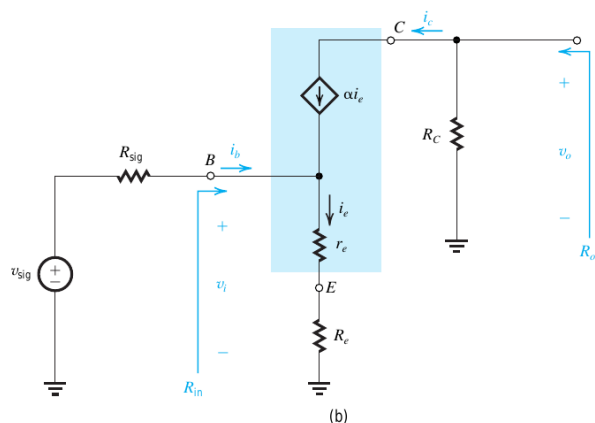
$$\circ \Rightarrow R_{in} = \frac{v_i}{i_b}$$

$$\blacksquare \text{ As } i_b = \frac{i_e}{\beta+1} \Rightarrow R_{in} = \frac{v_i}{i_b} = \frac{(\beta+1)v_i}{i_e}$$

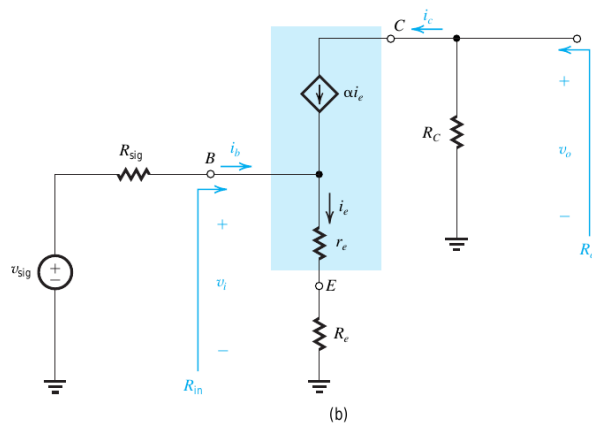
$$\blacksquare \text{ As } v_i \text{ appears across } r_e + R_e \Rightarrow \text{by ohms law}$$

$$\blacksquare i_e = \frac{v_i}{r_e + R_e}$$

$$\blacksquare \Rightarrow R_{in} = \frac{(\beta+1)v_i}{i_e} = \frac{(\beta+1)v_i}{\frac{v_i}{r_e + R_e}} = (\beta + 1)(r_e + R_e)$$



- $\circ \Rightarrow$ the input resistance looking into the base is $(\beta + 1)$ times the resistance in the emitter
- the input resistance looking into the base is $(\beta + 1)$ times the resistance in the emitter
 - $\circ$ this is known as resistance-reflection rule
 - $\circ$ Note that here the factor $(\beta + 1)$ arises because the base current is $1/(\beta + 1)$ times the emitter current



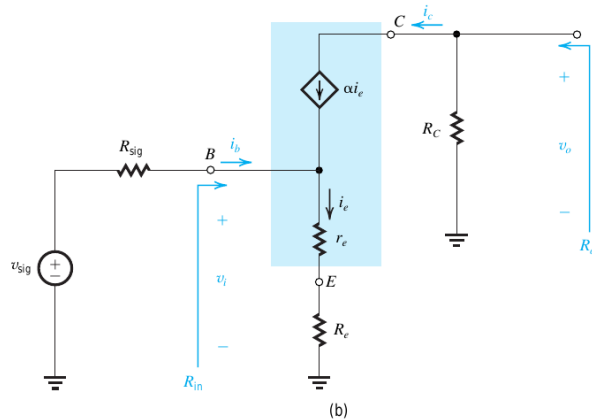
- it is clear from the above expression for R_{in} , that the input resistance has increased when R_e is included

o i.e. $\frac{R_{in}(\text{with } R_e \text{ included})}{R_{in}(\text{without } R_e)} = \frac{(\beta+1)(r_e+R_e)}{(\beta+1)r_e} = \frac{(r_e+R_e)}{r_e} = 1 + \frac{R_e}{r_e}$

o As $r_e = \frac{\alpha}{g_m}$

■ $\Rightarrow \frac{R_{in}(\text{with } R_e \text{ included})}{R_{in}(\text{without } R_e)} = 1 + \frac{R_e}{r_e} = 1 + \frac{g_m R_e}{\alpha} \approx 1 + g_m R_e : \alpha \approx 1$

■ $\Rightarrow$ As a circuit designer, we can use the value of R_e to control the value of R_{in}



- Now to determine voltage gain A_{vo}

- o KCL at C

■ $\Rightarrow \frac{v_o}{R_C} + \alpha i_e = 0$ or $\frac{v_o}{R_C} = -\alpha i_e$

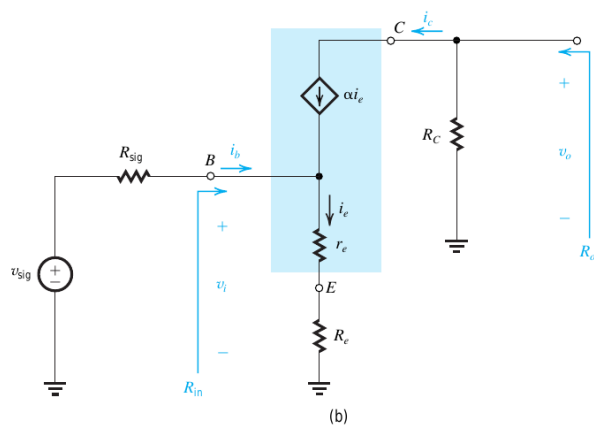
■ $\Rightarrow v_o = -\alpha i_e R_C$

- by ohm's law

■ $i_e = \frac{v_i}{r_e + R_e}$

■ $\Rightarrow v_o = -\alpha R_C \frac{v_i}{r_e + R_e}$

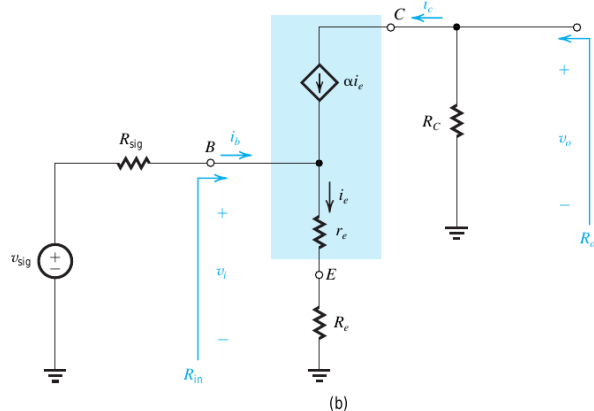
■ or $A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L=\infty} = -\alpha \frac{R_C}{r_e + R_e}$



$$\circ \text{ using } r_e = \frac{\alpha}{g_m} \Rightarrow A_{vo} = \frac{-\alpha R_C}{r_e + R_e} = \frac{-\alpha R_C}{\frac{\alpha}{g_m} + R_e} = \frac{-\alpha g_m R_C}{\alpha + g_m R_e} = \frac{-g_m R_C}{1 + \frac{g_m R_e}{\alpha}}$$

$$\bullet A_{vo} = \frac{-g_m R_C}{1 + \frac{g_m R_e}{\alpha}}$$

$$\circ A_{vo} \approx \frac{-g_m R_C}{1 + g_m R_e} \because \alpha \approx 1$$



◦ thus including R_e reduces the voltage gain by the factor $(1 + g_m R_e)$

◦ while it increases R_{in} by the same factor $\because \frac{R_{in}(\text{with } R_e \text{ included})}{R_{in}(\text{without } R_e)} \approx 1 + g_m R_e$

• the output resistance R_o can be determined by placing a test source at the collector

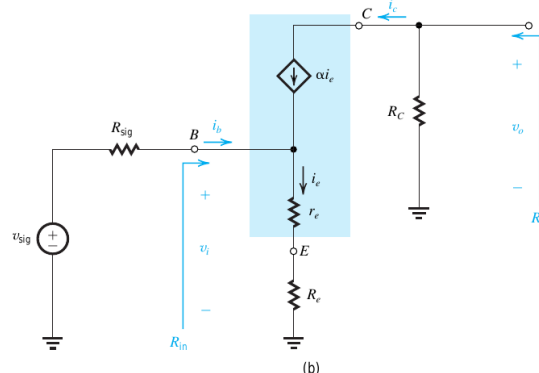
◦ and by setting $v_i = 0 \Rightarrow i_e = \frac{v_i}{r_e + R_e} = 0 \Rightarrow \alpha i_e = 0$

◦ $\alpha i_e = 0 \Rightarrow R_o = R_C$

■ now when R_L is connected at the output, A_v can be given as

$$\bullet A_v = A_{vo} \frac{R_L}{R_L + R_o}$$

■ As $A_{vo} = -\alpha \frac{R_C}{r_e + R_e}$, $R_o = R_C$



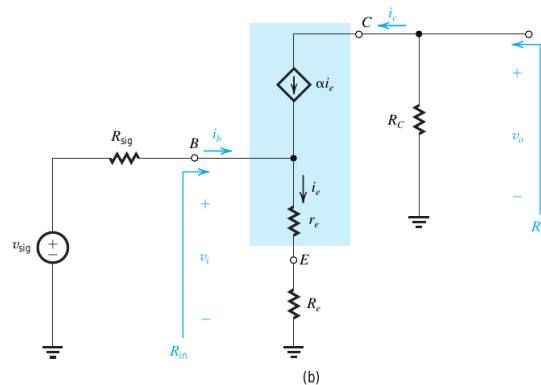
$$\circ \Rightarrow A_v = A_{vo} \frac{R_L}{R_L + R_o} = \left(-\alpha \frac{R_C}{r_e + R_e} \right) \frac{R_L}{R_L + R_C}$$

- $A_v = \left(-\alpha \frac{R_C}{r_e + R_e}\right) \frac{R_L}{R_L + R_C} = \left(-\frac{\alpha}{r_e + R_e}\right) \frac{R_C R_L}{R_L + R_C}$

- $A_v = \left(-\frac{\alpha}{r_e + R_e}\right) \left(\frac{1}{\frac{1}{R_C} + \frac{1}{R_L}}\right) = -\frac{\alpha(R_C \parallel R_L)}{r_e + R_e}$

- $A_v = -\frac{\alpha(R_C \parallel R_L)}{r_e + R_e}$

- Note that A_v is $-\alpha$ times the ratio of the total resistance in collector to the total resistance in emitter

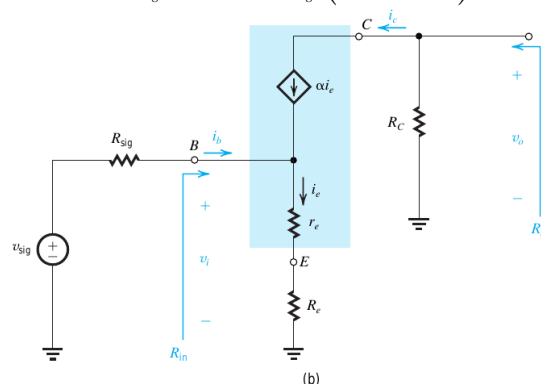


- $A_v = -\frac{\alpha(R_C \parallel R_L)}{r_e + R_e}$

- Now the overall gain G_v can be given as

- $G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v = \frac{R_{in}}{R_{in} + R_{sig}} A_v$

- $G_v = \frac{R_{in}}{R_{in} + R_{sig}} A_v = \frac{R_{in}}{R_{in} + R_{sig}} \left(-\frac{\alpha(R_C \parallel R_L)}{r_e + R_e}\right)$



- As $R_{in} = (\beta + 1)(r_e + R_e)$

- $\Rightarrow G_v = \frac{(\beta + 1)(r_e + R_e)}{(\beta + 1)(r_e + R_e) + R_{sig}} \left(-\frac{\alpha(R_C \parallel R_L)}{r_e + R_e}\right)$

- $G_v = \frac{(\beta + 1)(r_e + R_e)}{(\beta + 1)(r_e + R_e) + R_{sig}} \left(-\frac{\alpha(R_C \parallel R_L)}{r_e + R_e}\right)$

- As $\alpha = \frac{\beta}{\beta + 1} \Rightarrow \alpha(\beta + 1) = \beta$

- $G_v = \frac{\beta(r_e + R_e)}{(\beta + 1)(r_e + R_e) + R_{sig}} \left(-\frac{(R_C \parallel R_L)}{r_e + R_e}\right) = \frac{-\beta(R_C \parallel R_L)}{(\beta + 1)(r_e + R_e) + R_{sig}}$

- $G_v = -\beta \frac{R_C \parallel R_L}{(\beta + 1)(r_e + R_e) + R_{sig}}$

The Common-Base (CB) Amplifier

- here common-base amplifier is shown

- Note that the biasing circuit is not shown in figure

- the amplifier is fed with a signal source characterized by v_{sig} and R_{sig}

- Note that R_{sig} appears in series with the source

- $\Rightarrow$ it is more convenient to use the T model rather than the π model

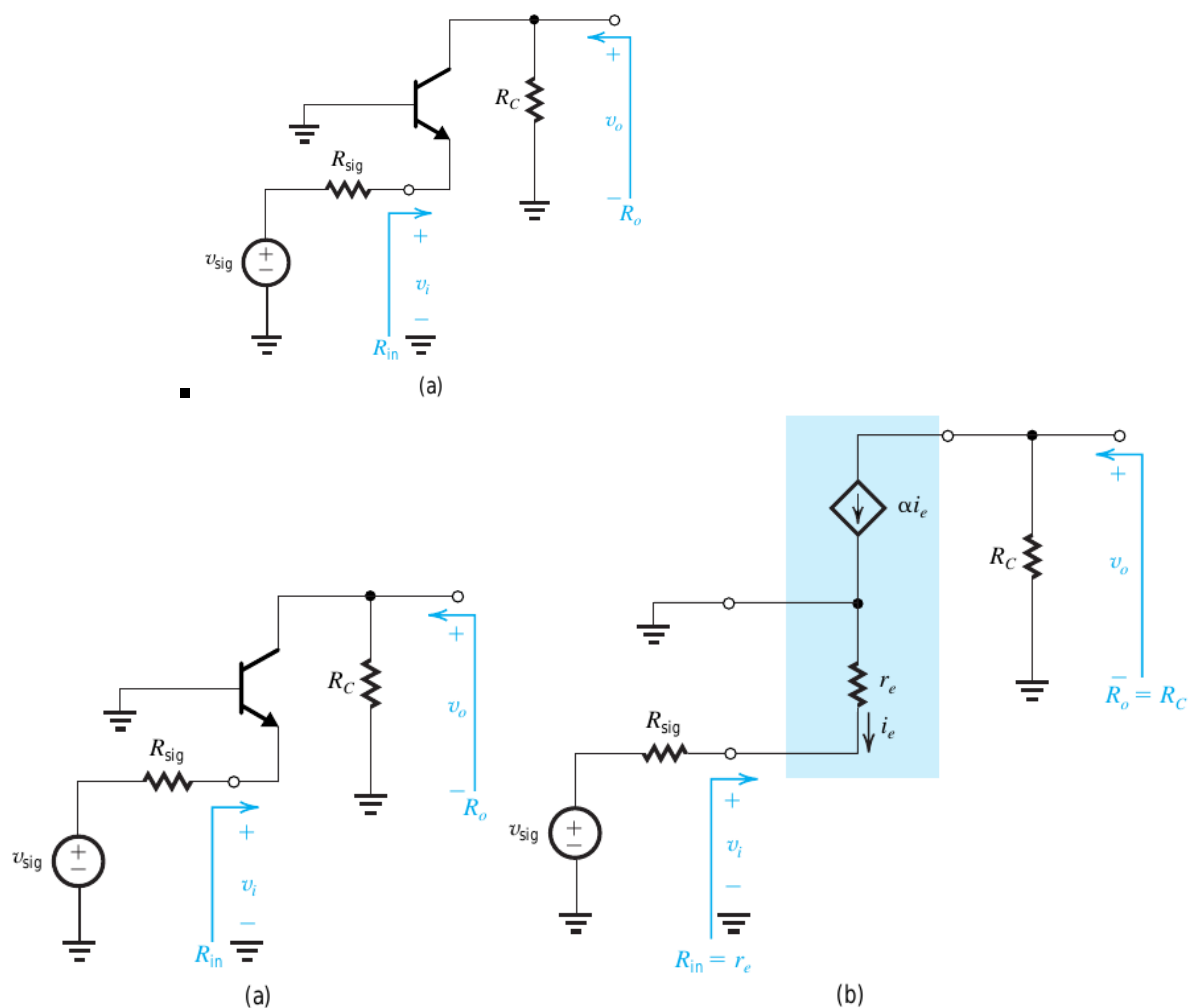
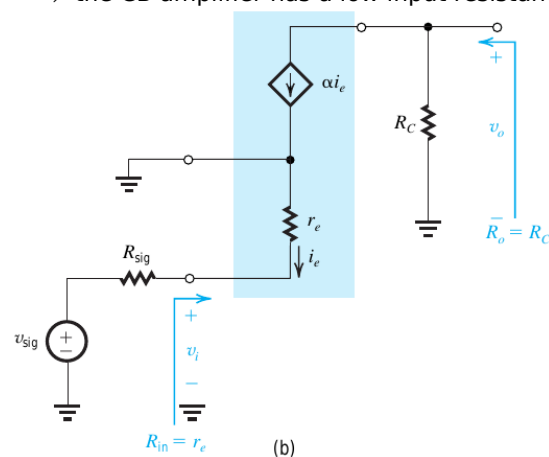


Figure 6.53 (a) CB amplifier with bias details omitted; (b) Amplifier equivalent circuit with the BJT represented by its T Model.

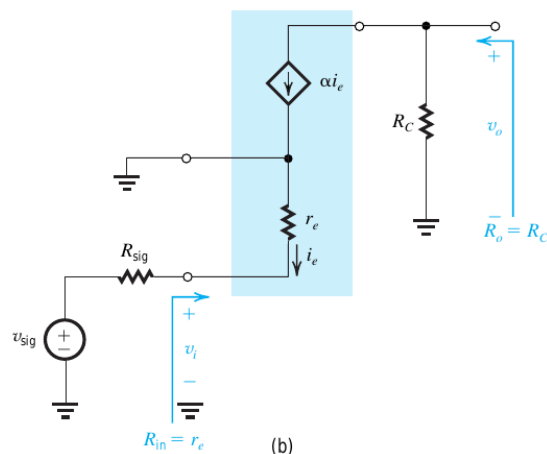
- here Early Effect has been ignored i.e. $r_o = \infty$
- the input resistance R_{in} can be given as $R_{in} = \frac{v_i}{i_i} = \frac{-v_{be}}{-i_e} = \frac{v_{be}}{i_e} = r_e$
 - as R_{in} is the resistance looking into the emitter and the base is grounded
 - typically r_e is a few ohms to a few tens of ohms

■ $\Rightarrow$ the CB amplifier has a low input resistance



- Now to determine the voltage gain A_{vo}
 - by KCL at C
 - $\frac{v_o}{R_C} + \alpha i_e = 0 \Rightarrow v_o = -\alpha R_C i_e$
 - $v_o = -\alpha R_C i_e$

■ where i_e is the emitter current i.e. $i_e = \frac{0-v_i}{r_e} = \frac{-v_i}{r_e}$



$$\circ \Rightarrow v_o = -\alpha R_C i_e = -\alpha R_C \left(\frac{-v_i}{r_e} \right) = \frac{\alpha R_C v_i}{r_e} \Rightarrow A_{vo} = \frac{v_o}{v_i} = \alpha \frac{R_C}{r_e} = g_m R_C \cdot r_e = \frac{\alpha}{g_m}$$

- The output resistance of the CB amplifier is determined by applying a test source at the output and setting $v_i = 0$

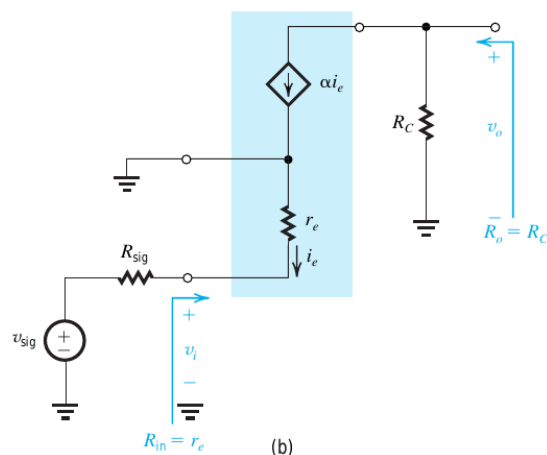
$$\circ v_i = 0 \Rightarrow i_e = \frac{-v_i}{r_e} = 0 \Rightarrow \alpha i_e = 0$$

$$\circ \Rightarrow R_o = R_C$$

○ the gain A_v is

$$\blacksquare A_v = \frac{v_o}{v_i} = g_m (R_C \parallel R_L)$$

$$\blacksquare \because A_{vo} = g_m R_C \text{ and as } R_L \text{ is in parallel with } R_C$$



- To determine the overall gain G_v

○ first by voltage divider

$$\blacksquare v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig} = \frac{r_e}{r_e + R_{sig}} v_{sig} \because R_{in} = r_e$$

$$\blacksquare G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i}$$

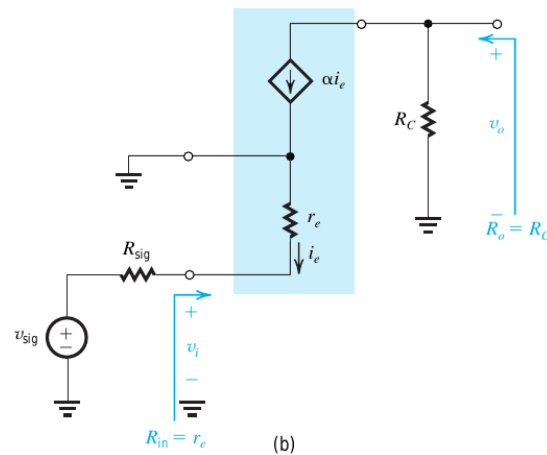
$$\blacksquare G_v = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v = \frac{r_e}{r_e + R_{sig}} A_v$$

$$\blacksquare \because A_v = g_m (R_C \parallel R_L)$$

$$\blacksquare G_v = \frac{r_e}{r_e + R_{sig}} g_m (R_C \parallel R_L)$$

$$\blacksquare \text{ As } r_e = \frac{\alpha}{g_m} \Rightarrow r_e g_m = \alpha$$

$$\blacksquare G_v = \frac{\alpha}{r_e + R_{sig}} (R_C \parallel R_L)$$

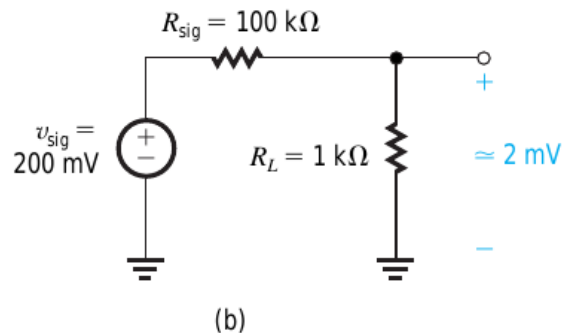
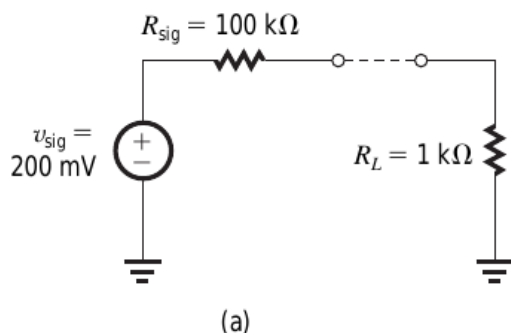


- $G_v = \frac{\alpha}{r_e + R_{sig}} (R_C \parallel R_L) = \alpha \frac{(R_C \parallel R_L)}{r_e + R_{sig}}$
 - $G_v \approx \frac{(R_C \parallel R_L)}{r_e + R_{sig}} \because \alpha \approx 1$
 - thus the overall voltage gain is simply the ratio of
 - the total resistance in the collector circuit
 - to the total resistance in the emitter circuit
 - CB amplifier has excellent frequency response and can be used to
 - amplify high-frequency signals that originate from sources
 - with relatively low resistances

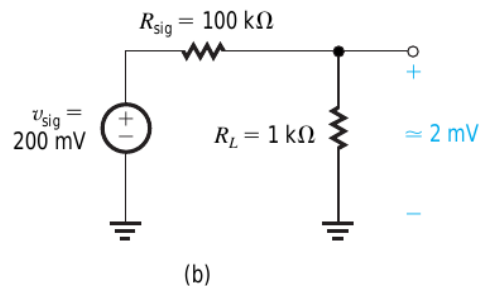
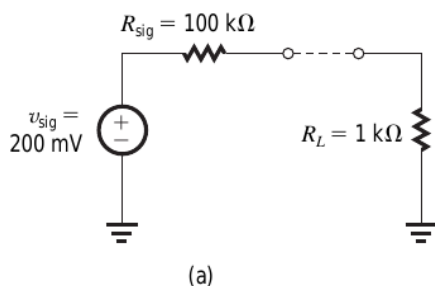
The Common-Collector Amplifier or Emitter Follower

- The common-collector amplifier is more commonly known as the emitter follower
 - A common application of a CC amplifier is as a voltage buffer

The need for Voltage Buffers



- - consider the situation shown in fig
 - here a signal source with open-circuit voltage of $v_{sig} = 200\text{mV peak}$
 - and internal resistance of $100\text{k}\Omega$ is to be connected to a $1\text{k}\Omega$ load resistance

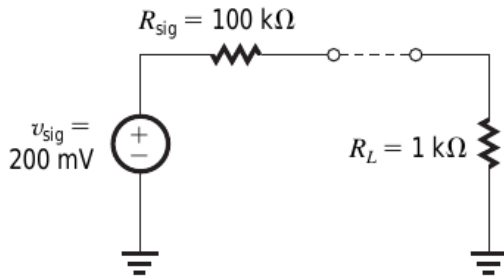


- - connecting the source to the load directly will result in
 - severe attenuation of the signal
 - i.e. the voltage appearing across the load is

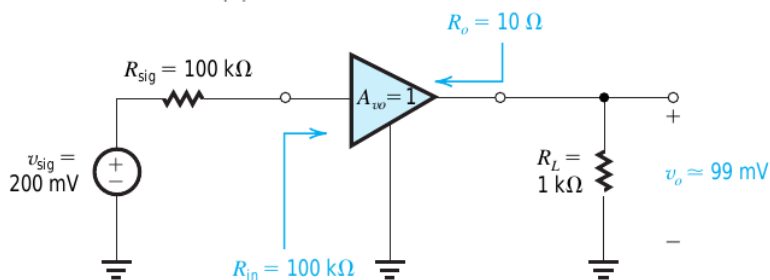
$$\blacksquare v_o = \frac{R_L}{R_L + R_{sig}} v_{sig} = \frac{1k}{1k + 100k} v_{sig}$$

$$\blacksquare \text{As } v_{sig} = 200mV \Rightarrow v_o = \frac{1k}{1k + 100k} v_{sig} = 1.98mV \approx 2mV$$

$\blacksquare$ thus for $v_{sig} = 200mV$ peak, output voltage is only $v_o = 2mV$ peak

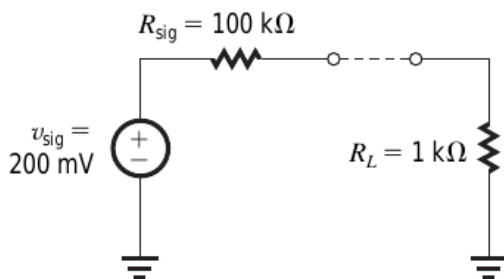


(a)

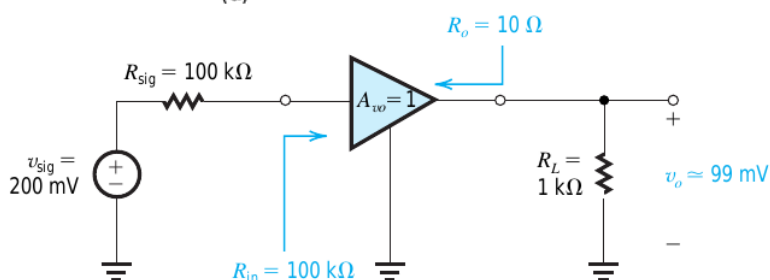


(c)

- now suppose, we have an amplifier as shown in fig
 - this amplifier is used to connect the source to the load
 - $\blacksquare$ this amplifier has a unity voltage gain
 - $\blacksquare$ (here our signal is already of sufficient strength and increase in its amplitude is not required)

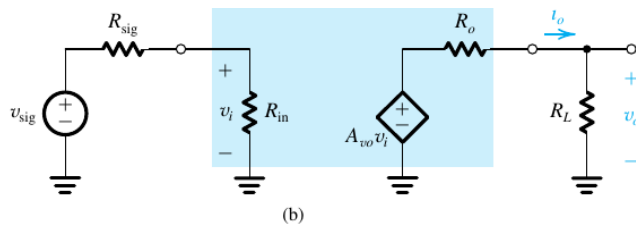
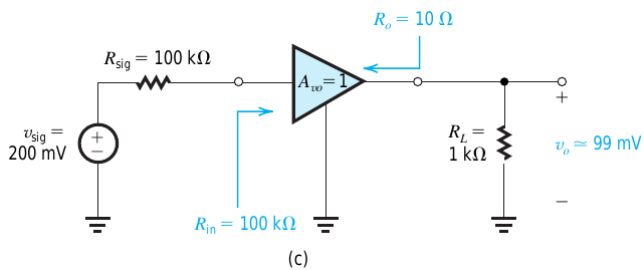


(a)



(c)

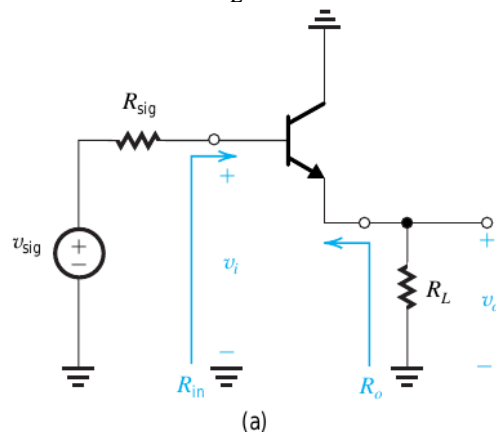
- this amplifier has an input resistance, $R_{in} = 100k\Omega$
 - $\Rightarrow v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig} = \frac{1}{2} v_{sig}$
 - $\blacksquare$ thus for $v_{sig} = 200mV$, $v_i = 100mV$



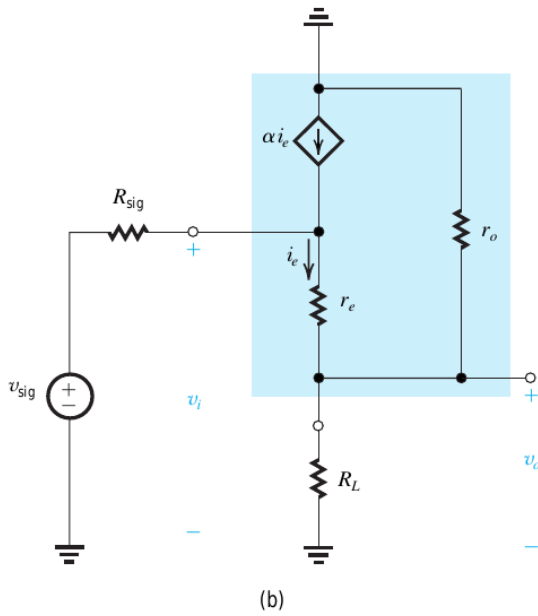
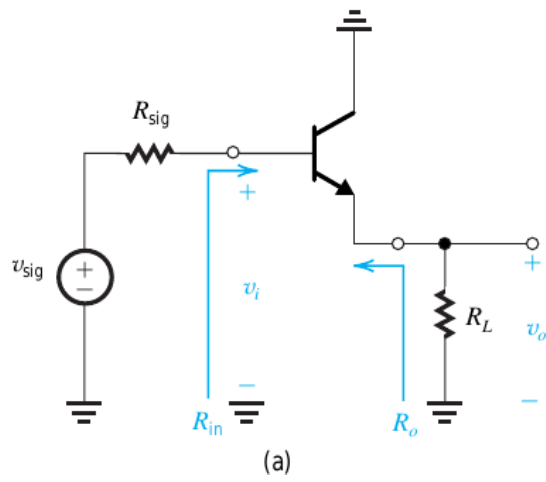
- the amplifier has a low output resistance of $R_o = 10\Omega$
 - here $R_o = 10\Omega$, $R_L = 1k\Omega$, $v_i = \frac{v_{sig}}{2}$, $A_{vo} = 1$
 - by voltage divider
 - $v_o = \frac{R_L}{R_L + R_o} A_{vo} v_i = \frac{1k}{1k + 10} (1) v_i = 0.9901 v_i$
 - $\Rightarrow 99\%$ of v_i appears at the output
 - and $v_o = \frac{0.9901}{2} v_{sig}$
 - this is a significant improvement over the situation with the source connected directly to the load.
 - such an amplifier circuit with $A_{vo} = 1$, $R_{in} \rightarrow large$ and $R_{out} \rightarrow small$ is called a voltage buffer

The Common-Collector Amplifier or Emitter Follower

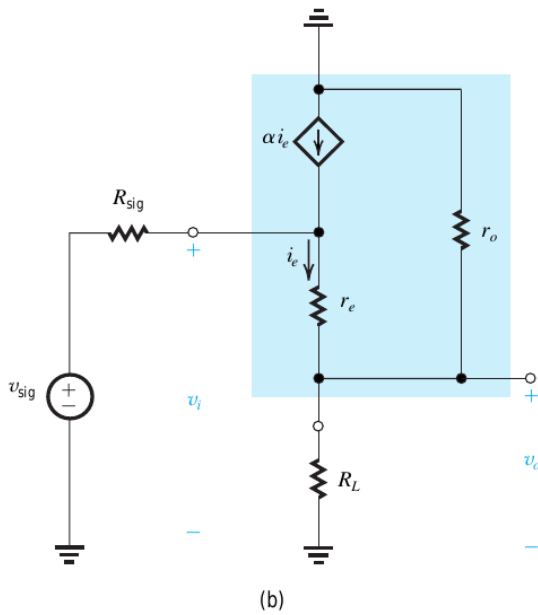
- the CC amplifier with the bias circuit omitted is shown here
 - a signal generator (v_{sig} , R_{sig}) is applied between the base and ground
 - a load resistance R_L is connected between the emitter terminal and ground

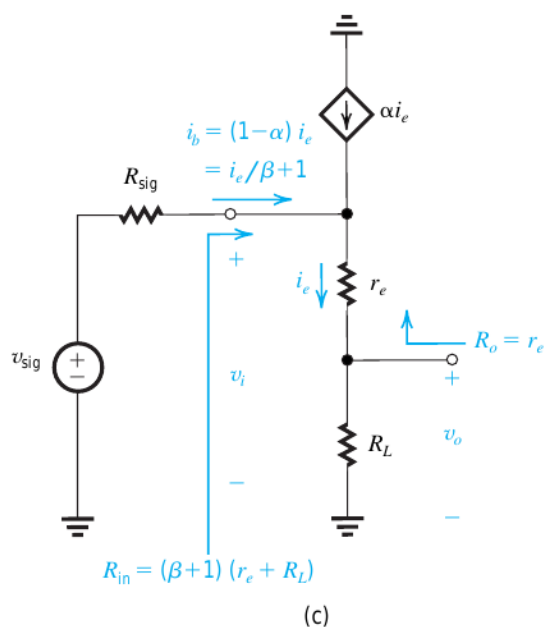


- As R_L is connected to the emitter terminal, it is more convenient to use the T-model rather than the hybrid- π model
- here r_o appears in parallel with R_L . so base-width modulation effect can easily be incorporated.



- if we ignore base-width modulation $\Rightarrow r_o = \infty$

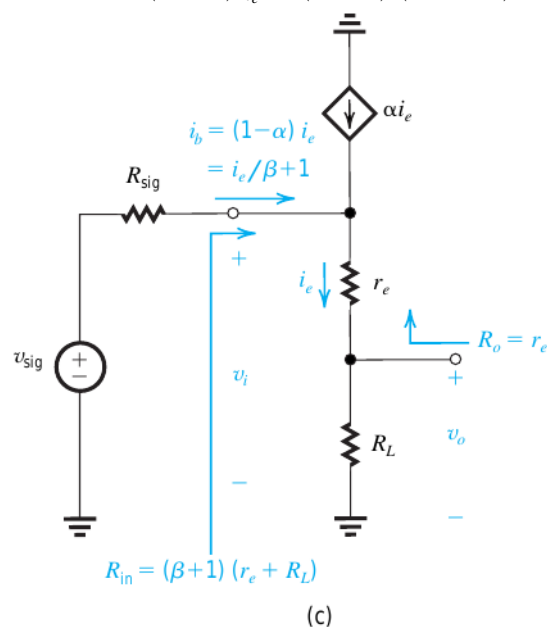




- from figure, the input resistance R_{in} can be given as

$$R_{in} = \frac{v_i}{i_b}$$

- using KCL $i_b + \alpha i_e = i_e \Rightarrow i_b = i_e(1 - \alpha)$
- $i_b = i_e \left(1 - \frac{\beta}{\beta+1} \right) = i_e \left(\frac{\beta+1-\beta}{\beta+1} \right) = \frac{i_e}{\beta+1}$
- $\Rightarrow R_{in} = \frac{v_i}{i_b} = (\beta+1) \frac{v_i}{i_e}$
- As by ohms law $i_e = \frac{v_i}{r_e + R_L} \Rightarrow r_e + R_L = \frac{v_i}{i_e}$
- $\Rightarrow R_{in} = (\beta+1) \frac{v_i}{i_e} = (\beta+1)(r_e + R_L)$

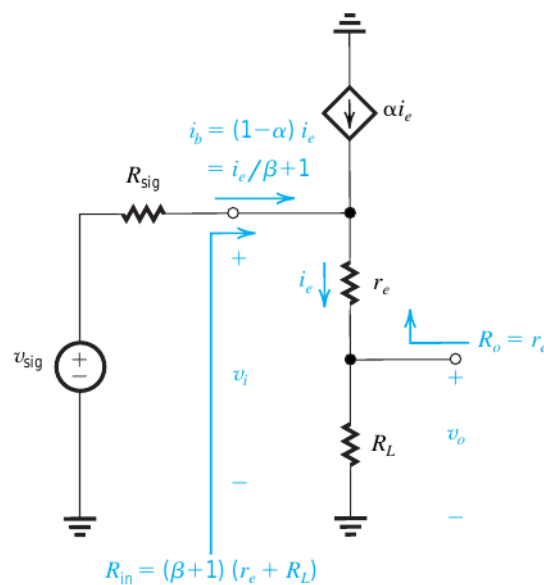


- The voltage gain A_v is given as

$$A_v = \frac{v_o}{v_i}$$

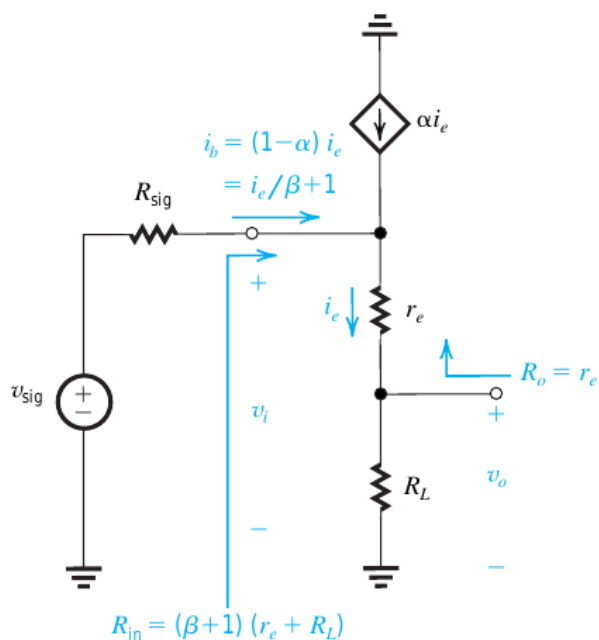
- Apply KCL at node v_o
- $\Rightarrow i_e = \frac{v_o}{R_L}$
- As $i_e = \frac{v_i - v_o}{r_e}$
- $\Rightarrow \frac{v_i - v_o}{r_e} = \frac{v_o}{R_L}$

- $\frac{v_i}{r_e} = \frac{v_o}{R_L} + \frac{v_o}{r_e}$ or $\frac{v_i}{r_e} = v_o \left(\frac{1}{R_L} + \frac{1}{r_e} \right)$
- $v_o \left(\frac{1}{R_L} + \frac{1}{r_e} \right) = \frac{v_i}{r_e} \Rightarrow v_o \left(\frac{r_e + R_L}{R_L r_e} \right) = \frac{v_i}{r_e}$
- $v_o \left(\frac{r_e + R_L}{R_L} \right) = v_i$



(c)

- $v_o \left(\frac{r_e + R_L}{R_L} \right) = v_i$
 - $\Rightarrow \frac{v_o}{v_i} = \frac{R_L}{r_e + R_L}$
 - thus $A_v = \frac{v_o}{v_i} = \frac{R_L}{r_e + R_L}$
 - and A_{vo} can be given as
 - $A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L=\infty} = \left. \frac{R_L}{r_e + R_L} \right|_{R_L=\infty}$
 - $A_{vo} = \left. \frac{1}{\frac{r_e}{R_L} + 1} \right|_{R_L=\infty} = \frac{1}{0+1} = 1$

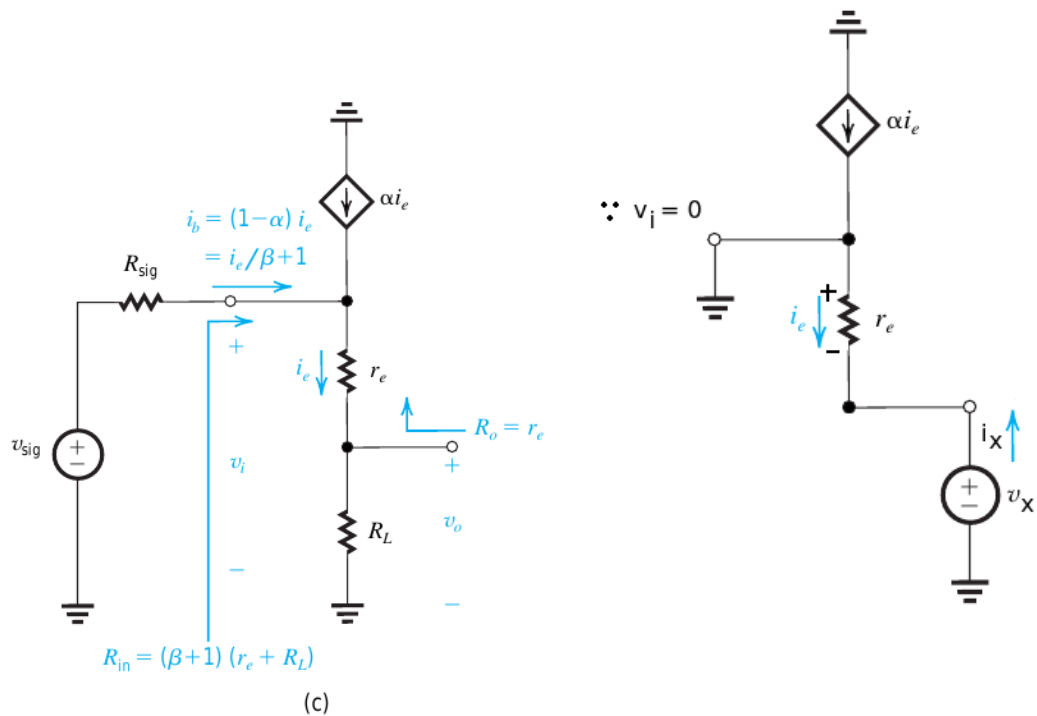


(c)

- the output resistance R_o is determined by removing R_L and looking into the emitter while setting

$$v_i = 0$$

◦ using a test source v_x with current $i_x \Rightarrow R_o = \frac{v_x}{i_x}$



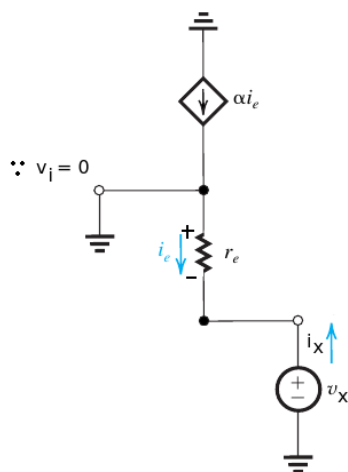
• here base terminal is grounded $\therefore v_i = 0$

◦ Ohm's Law $\Rightarrow i_e = \frac{0-v_x}{r_e} = -\frac{v_x}{r_e}$

$$\blacksquare \Rightarrow i_x = -i_e = \frac{v_x}{r_e}$$

$$\blacksquare \text{ or } i_x r_e = v_x \Rightarrow \frac{v_x}{i_x} = r_e$$

$$\blacksquare \Rightarrow R_o = \frac{v_x}{i_x} = r_e$$



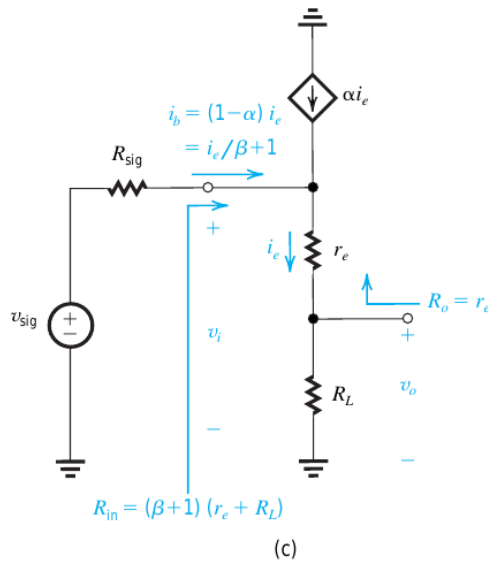
$$\bullet G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v$$

◦ using voltage divider rule, $v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig}$

$$\blacksquare G_v = \frac{v_i}{v_{sig}} A_v = \frac{R_{in}}{R_{in} + R_{sig}} A_v$$

$$\blacksquare \text{ As } R_{in} = (\beta + 1)(r_e + R_L), A_v = \frac{R_L}{r_e + R_L}$$

$$\begin{aligned} \blacksquare G_v &= \frac{R_{in}}{R_{in} + R_{sig}} A_v = \frac{(\beta+1)(r_e + R_L)}{(\beta+1)(r_e + R_L) + R_{sig}} \left(\frac{R_L}{r_e + R_L} \right) \\ \blacksquare G_v &= \frac{(\beta+1)R_L}{(\beta+1)(r_e + R_L) + R_{sig}} \end{aligned}$$



- $G_v = \frac{(\beta+1)R_L}{(\beta+1)(r_e + R_L) + R_{sig}}$
 - thus the overall gain, though lower than one, can be close to one if $(\beta+1)R_L$ is larger or comparable in value to R_{sig}
 - i.e. R_L can be much smaller than R_{sig}

Basic BJT Amplifier Configurations

TABLE 6.5 Characteristics of BJT Amplifiers^{a, b, c}

	R_{in}	A_{vo}	R_o	A_v	G_v
Common emitter (Fig. 6.50)	$(\beta+1)r_e$	$-g_m R_C$	R_C	$-g_m(R_C \parallel R_L)$ $-\alpha \frac{R_C \parallel R_L}{r_e}$	$-\beta \frac{R_C \parallel R_L}{R_{sig} + (\beta+1)r_e}$
Common emitter with R_e (Fig. 6.52)	$(\beta+1)(r_e + R_e)$	$-\frac{g_m R_C}{1 + g_m R_e}$	R_C	$\frac{-g_m(R_C \parallel R_L)}{1 + g_m R_e}$ $-\alpha \frac{R_C \parallel R_L}{r_e + R_e}$	$-\beta \frac{R_C \parallel R_L}{R_{sig} + (\beta+1)(r_e + R_e)}$
Common base (Fig. 6.53)	r_e	$g_m R_C$	R_C	$g_m(R_C \parallel R_L)$ $\alpha \frac{R_C \parallel R_L}{r_e}$	$\alpha \frac{R_C \parallel R_L}{R_{sig} + r_e}$
Emitter follower (Fig. 6.55)	$(\beta+1)(r_e + R_L)$	1	r_e	$\frac{R_L}{R_L + r_e}$	$\frac{R_L}{R_L + r_e + R_{sig}/(\beta+1)}$ $G_{vo} = 1$ $R_{out} = r_e + \frac{R_{sig}}{\beta+1}$

^a For the interpretation of R_{in} , A_{vo} , and R_o refer to Fig. 6.49.^b The BJT output resistance r_o has been neglected, which is permitted in the discrete-circuit amplifiers studied in this chapter. For integrated-circuit amplifiers (Chapter 7), r_o must always be taken into account.^c Setting $\beta = \infty$ ($\alpha = 1$) and replacing r_e with $1/g_m$, R_C with R_D , and R_e with R_s results in the corresponding formulas for MOSFET amplifiers (Table 5.4).